

Project #8: AUTONOMOUS MOBILE MANIPULATOR ROBOT

Indoor Navigation and Smart Object Handling Across Building Floors

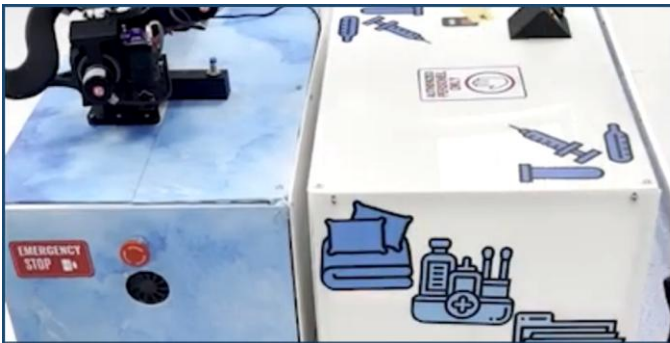
PROJECT TYPE Graduation Project	CORE PLATFORM Mobile Base + 5/6-DOF Arm	AUTONOMY SLAM • Perception • Planning	TARGET Multi-Floor Smart Delivery
 Existing Mobile-Manipulator Prototype Integrated base, arm, emergency stop and payload compartment		 Door-Handle Interaction Demonstration Manipulator reaches and operates a human-scale door handle	

Figure 1. Existing mobile-manipulator prototype and demonstrated door-handle interaction used as the development baseline.

AUTONOMOUS MOBILE MANIPULATOR

Multi-floor Delivery for Hospitals & Smart Buildings

1 Autonomous Navigation



2 Elevator Interaction



3 Door Handling



4 Delivery at Destination



-  Carries Medical Supplies & Small Packages
-  Autonomous Navigation & Obstacle Avoidance
-  Elevator Interaction & Multi-floor Operation
-  Autonomous Door Handling & Access
-  Safe, Reliable & Suitable for Hospitals & Smart Buildings





MULTI-FLOOR DELIVERY

Autonomous. Reliable. Efficient.

From pickup to drop-off across multiple floors, our mobile manipulator robot streamlines internal logistics and enhances operational efficiency.

1. Project Description

This graduation project develops and deploys a fully autonomous mobile manipulator robot that can accept a user request, plan a safe route, navigate through indoor corridors, interact with doors and elevators, grasp or transport small objects, and complete service missions on different building floors. The platform integrates a four-wheel hub-motor mobile base, a 5/6-DOF robotic arm with a suitable gripper, onboard computing, perception sensors, wireless communication and an intelligent battery-management system. Building on the previous-year prototype, the new work focuses on reliable autonomy in human-populated environments and coordinated control between locomotion and manipulation. The robot will be designed for hospitals, offices, laboratories and academic buildings, where it can deliver medicines, documents, tools or small packages while providing live status to authorized users through a mobile application.

2. Motivation, Impact and Main Objectives

Service operations across large buildings are repetitive, time-consuming and frequently require staff to move items between distant rooms or floors. A mobile manipulator can reduce this burden, support contactless delivery, improve response time and provide safer assistance during emergencies or periods of high workload. The project also provides students with an integrated challenge covering mechanical design, embedded control, autonomous navigation, computer vision, robotic manipulation, networking, software engineering and safety validation.

- Upgrade the mobile platform, hub-motor drives, odometry and mechanical integration for stable indoor operation.
- Implement LiDAR/camera/IMU-based SLAM, localization, path planning and dynamic obstacle avoidance.
- Develop precise arm and gripper control for grasping, button pressing and door-handle operations.
- Achieve autonomous elevator calling, entry, floor selection, exit and floor-map transition.
- Create a secure cross-platform mobile application for task assignment, tracking and notifications.
- Integrate BMS monitoring, automatic docking, fault handling, emergency stop and human-aware safety.

3. Integrated System Architecture and Technical Modules

The platform is organized as a layered architecture so that mission-level planning, perception, real-time motion control, safety-critical functions, communication and energy management can be developed and tested independently before full-system integration. Sensor fusion provides a shared estimate of robot pose, obstacles, people, doors, elevator panels and manipulated objects. The mission manager selects the appropriate navigation or manipulation behavior and supervises recovery when the environment changes.

Integrated Autonomous Mobile-Manipulation Architecture

Mission-level autonomy, perception, real-time control, safety, communication and energy management

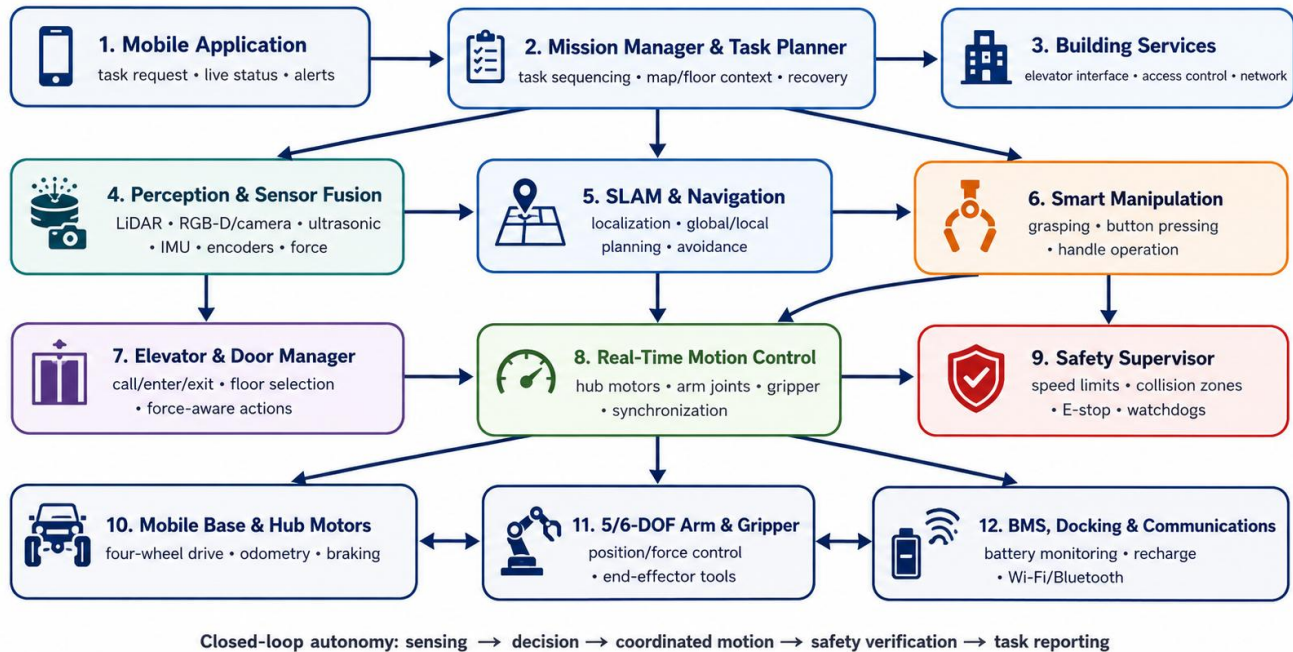


Figure 2. Proposed system architecture linking user requests, autonomous navigation, smart manipulation, safety, actuators, communication and power management.

Mobile Platform and Drive	Four-wheel hub-motor locomotion with motor drivers, encoder odometry, braking, speed regulation and a mechanically stable arm mounting structure. The base must negotiate corridors, doorways and elevator thresholds.
Perception and Localization	LiDAR, RGB-D or monocular cameras, ultrasonic/proximity sensors, IMU and wheel encoders are fused for mapping, localization, obstacle detection, human awareness and semantic recognition of doors, panels and objects.
SLAM and Motion Planning	A navigation stack generates global and local paths, applies dynamic obstacle avoidance, switches between floor maps and supports precise alignment at doors, elevators, charging stations and delivery positions.
Manipulator and End Effector	The 5/6-DOF arm uses joint position/velocity control with optional force or current feedback. Motion planning, inverse kinematics and grasp strategies enable button pressing, handle rotation and safe object transfer.
Elevator and Door Interaction	Computer vision identifies controls and handles; guarded force-aware motions verify contact. The robot calls the elevator, enters, selects the target floor, detects arrival and exits while maintaining localization.
Task App, Networking and Data	A secure mobile app assigns missions and displays location, battery and status. Wi-Fi/Bluetooth supports telemetry, remote diagnostics, event logs and connection to building services or a supervisory dashboard.
Energy and Safety Management	The BMS monitors state of charge, current, temperature and faults. Low-energy missions trigger automatic docking. Emergency stop, watchdogs, speed zones, collision limits and safe recovery protect people and hardware.

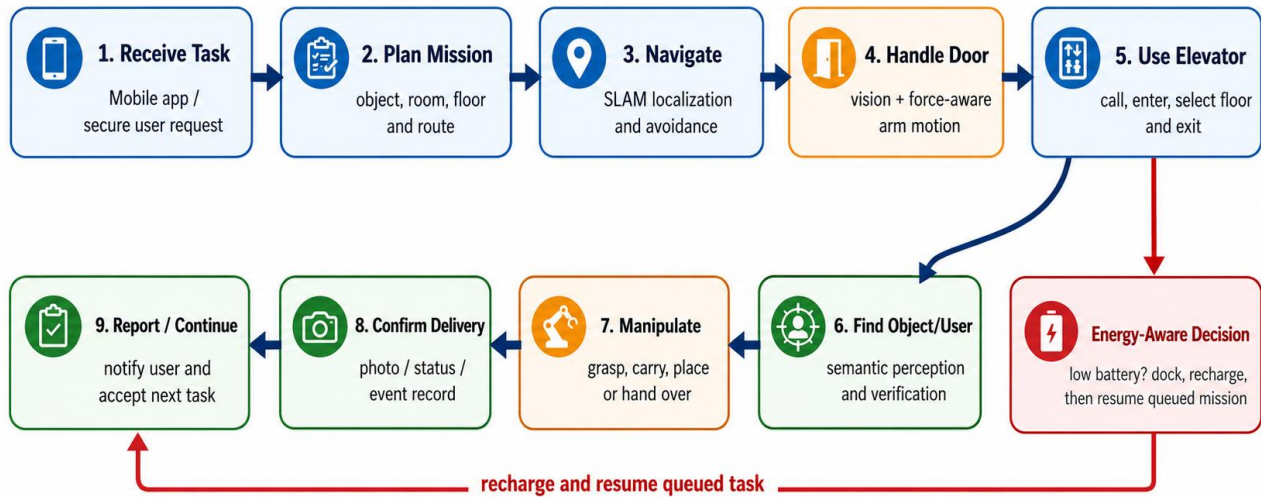
4. Control and Software Framework

The software will use a modular robotics framework such as ROS/ROS 2 to connect perception, SLAM, navigation, manipulation, elevator interaction, mobile-app communication and diagnostics. A behavior tree or finite-state mission manager will sequence actions and handle timeouts, blocked paths, failed grasps, unavailable elevators and communication loss. Low-level wheel and arm loops run at deterministic rates, while higher-level planners update paths and actions from live sensor data. Data logging will store trajectories, images, forces, battery information, mission events and safety interventions for debugging and quantitative evaluation.

5. Autonomous Mission Sequence, Development Plan and Validation

A complete mission begins with a secure request and ends only after delivery verification, user notification and safe continuation or recharging. The workflow below emphasizes multi-floor navigation, coordinated door/elevator manipulation and energy-aware task resumption.

End-to-End Multi-Floor Autonomous Mission Workflow



At every stage: human detection, speed limitation, collision checking, emergency stop and fault recovery remain active.

Figure 3. End-to-end autonomous mission sequence with safety monitoring and an energy-aware docking/resumption branch.

Applications: hospital logistics, office and laboratory delivery, smart-building services, contactless support and assistance during emergencies.

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Previous-year demonstration: [YouTube video](#)